

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering
Course Outline: Data Communication

Part A- Introduction

I. Course code: CSE 2201

II. Credit: 3

1. Course Summary

This course introduces some basic concepts, models, techniques etc. used in data communication. A computer science engineer needs to know the communication model, different network layer, transmission medium and technique used in digital communication system to fulfill his CSE degree.

2. Course Objectives

1. Build an understanding of the fundamental concepts of Data communication.
2. Learn how computer network hardware and software operate.
3. Investigate the fundamental issues driving network design.
4. Learn about dominant network technologies.

3. Course Learning Outcomes

- CLO1.** Learn about the basic knowledge of data communication systems
- CLO2.** Able to understand about various layered model used in data communication
- CLO3.** Explore different methods and techniques used in data transfer
- CLO4.** Learn about various Protocols and their packet format
- CLO5.** Understand the uses and effectiveness of various transmission medium

Part B- Lesson Plan

Course Details Learning plan

Timelin e	Topics /contents	Learning Outcomes	Mappe d CLOs	Teachin g Strategie s	Assessme nt Strategies
Week 1	Data Communication, Fundamental characteristics, Components, Data representation, Data Flow, Network and its criteria, point-to-point and multipoint connection, different topologies, Network Models, Protocols and standards	Learn about data Communication model, network standards and organizations, different protocol layers	CLO1, CLO4, CLO5	Lecture	Essay Short Question
Week 2	Different tasks in network model, peer-to-peer communication, characteristics and usage area of different layers in OSI model, fundamental properties of different layers of TCP/IP protocol, physical, logical and port address.	Learn about Network Model, Tasks, ISO, OSI Model, TCP/IP Protocol	CLO1, CLO2, CLO4	Lecture	Quiz Assignme nt Essay Short Question
Week 3,4	Analog and digital data and signal, period, frequency, phase, wavelength, time and frequency domain, bandwidth, transmission of digital signal, attenuation, distortion, noise, SNR, Nyquist bit rate, Shannon's capacity performance parameters.	Learn about physical layer and media, Illustrate Analog and digital data and signal, transmission technique and its impairments, Data rate limit for both noisy and noiseless channel.	CLO1, CLO2, CLO3	Lecture Exercise Demonst ration	Quiz Assignme nt Essay Short Question
Week 5, 6	Line coding techniques NRZ, RZ, Manchester, and differential Manchester encoding, AMI, Block	Learn various Digital transmission techniques.	CLO3	Lecture Exercise	Quiz Essay Short

	coding, analog to digital conversion based on PCM, delta modulation, etc.				Question
Week 7	ASK, FSK, PSK, QPSK, QAM encodings, AM, PM, FM, etc.	Learn various Analog transmission techniques.	CLO3	Lecture Exercise	Quiz Essay Short Question
Week 8, 9	Synchronous and asynchronous data transmission techniques, FDM, international FDM carrier standards, synchronous TDM, international TDM carrier standards, statistical time division multiplexing	Discuss and characterize different data transmission and multiplexing techniques.	CLO3	Lecture Exercise	Assignment Essay Short Question
Week 10	Frequency hopping spread spectrum, direct sequence spread spectrum, code division multiple access.	Discuss and characterize different spread spectrum techniques.	CLO3	Lecture Exercise	Assignment Essay Short Question
Week 11, 12	Characteristics and applications of various types of guided medium, Characteristics and applications of wireless transmission-terrestrial and satellite microwave, radio waves, propagation mechanism, free space propagation, land propagation, path loss, slow fading, fast fading, delay spread, inter symbol interference, VSAT	Discuss and characterize different transmission medium, wireless transmission technique, Compute path loss, learn to distinguish slow and fast fading, Able to define inter symbol interference and VSAT	CLO5	Lecture Exercise	Assignment Essay Short Question

Week 13	Error Detection and Correction; parity check, CRC, forward error correction technique, linear block code, hamming code, etc.	Learn about error detection and correction methods, forward and backward detection, checksum	CLO1, CLO3	Lecture Exercise Demonstration	Assignment Essay Short Question
Week 14	Random access, ALOHA, Pure ALOHA, CSMA, CSMA/CD, CSMA/CA, Controlled access, reservation, token passing, polling, poll function, Channelization protocols, FDMA, TDMA and CDMA technology	Discuss wireless channel, multiple access protocol, random access and controlled access protocol	CLO3, CLO4	Lecture Exercise Demonstration	Assignment Essay Short Question

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	Description
Quiz, Short questions and Essay	These are simple class tests with duration from 10 minutes to 90 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of homework assessment, students are given specific tasks and instructed to complete them within a given period of time.
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3 hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	Test (10)	Assignment (5)	Quizzes (5)	External Participation in Curricular/Co-Curricular Activities
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	Test
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
CLO1	1	1	0	0	0	0	0	1	0
CLO2	1	1	0	0	0	0	0	1	1
CLO3	1	1	0	0	0	0	0	1	1
CLO4	1	1	0	0	0	0	0	1	1
CLO5	1	1	0	0	0	0	0	1	1

Mapping of CLOs to PLOs

Part D- Resources

Textbooks

1. Data Communications and networking by Behrouz A. Forouzan.
2. Data and Computer Communication by William Stallings.

Reference Books

1. Data Communication by Hajkins.
2. Data Communication by Taub.

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering

Course Outline: Database Management System

Part A- Introduction

I. Course code: CSE 2203

II. Credit: 3

1. Course Summary

A computer engineer needs to know the fundamentals of database architecture, database management systems, and database systems, principles and methodologies of database design, and techniques for database application development.

2. Course Objectives

1. An understanding of the needs for and uses of database management systems in business;
2. An understanding of the context, phases and techniques for designing and building database information systems in business;
3. An understanding of the components of a computerized database information system (application)
4. An ability to correctly use the techniques, components and tools of a typical database management system -- such as Access 2000 or Oracle 8i -- to build a comprehensive database information system (application);
5. An ability to design a correct, new database information system for a business functional area and implement the design in either Access 2000 or Oracle 8i;
6. An introductory understanding of some advanced topics in database management, e.g., object-relational databases and design, distributed databases, database administration (security, backup and restore, tuning) and data warehousing.

3. Course Learning Outcomes

- CLO1.** Database Concepts
- CLO2.** Relational Data Model
- CLO3.** Functional Dependency and Normal Forms
- CLO4.** Relational Calculus
- CLO5.** Relational Database Design
- CLO6.** Entity-Relationship (ER) approach
- CLO7.** Transformation of the ER model to SQL

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	Files and Databases, Database Management systems, Data models	Define Database Explain advantage of Database over file system Describe different data model	CLO 1	Lecture, handouts	Quiz Participation Exam Assignment
Week 2 ,3	Relations, Domains, Attributes and Tuple	Explain Relational Data model Apply RDM in DB design Describe different types of attribute	CLO 2	Lecture, handouts	Quiz Participation Exam Assignment

Week 4, 5	Anomalies, Functional Dependency, First, Second and third normal forms, Boyce-Codd Normal form	Evaluate anomalies in DB design Identify different normal form Apply normalization in DB design Convert one normal form to another normal	CLO 3	Lecture, handouts	Quiz Participation Exam Assignment
Week 6, 7, 8	Relational Calculus Based Languages: SQL, Relational algebra and Set operations	Understanding the concept of SQL	CLO 4	Lecture, handouts	Quiz Participation Exam Assignment
Week 9, 10	Relational design criteria, Lossless decomposition, decomposition algorithms, synthesis algorithms	Explain decomposition and synthesis algorithm Define lossless decomposition	CLO 5	Lecture, handouts	Quiz Participation Exam Assignment
Week 11, 12	The ER model and its constructs, ER modeling in logical database design.	Define entity Relationship, relationship set. Apply E-R approach to DB design	CLO 6	Lecture, handouts	Quiz Participation Exam Assignment
Week 13, 14	Transformation of the ER model to SQL	Explain transformation of ER model to SQL	CLO 7	Lecture, handouts	Quiz Participation Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	Description
Participation	Attendance and participate in the class discussion

Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	Test (10)	Assignment (5)	Quizzes (5)	External Participation in Curricular/Co-Curricular Activities
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	Test
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
CLO1	1	0	0	0	0	0	0	1	0
CLO2	1	0	0	0	0	0	0	1	0
CLO3	1	0	0	0	0	0	0	1	0
CLO4	1	0	0	0	0	0	0	1	0
CLO5	1	0	0	0	0	0	0	1	0
CLO6	1	0	0	0	0	0	0	1	0
CLO7	1	0	0	0	0	0	0	1	0

Part D- Resources

Textbooks

1. Database System Concepts by Korth and Silverchatz
2. Principle of Database Systems by O. William

Reference Books

1. Relational Database Management System by Jeffrey Ullman

Other resources (Slides)

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering

Course Outline: Database Management System Lab

Part A- Introduction

I. Course code: CSE 2204

II. Credit: 1.5

1. Course Summary

A computer engineer needs to know the fundamentals of database architecture, database management systems, and database systems, principles and methodologies of database design, and techniques for database application development.

2. Course Objectives

1. An understanding of the needs for and uses of database management systems in business;
2. An understanding of the context, phases and techniques for designing and building database information systems in business;
3. An understanding of the components of a computerized database information system (application)
4. An ability to correctly use the techniques, components and tools of a typical database management system -- such as Access 2000 or Oracle 8i -- to build a comprehensive database information system (application);
5. An ability to design a correct, new database information system for a business functional area and implement the design in either Access 2000 or Oracle 8i;
6. An introductory understanding of some advanced topics in database management, e.g., object-relational databases and design, distributed databases, database administration (security, backup and restore, tuning) and data warehousing.

3. Course Learning Outcomes

- CLO1.** Database Concepts
- CLO2.** Relational Data Model
- CLO3.** Functional Dependency and Normal Forms
- CLO4.** Relational Calculus
- CLO5.** Relational Database Design
- CLO6.** Entity-Relationship (ER) approach
- CLO7.** Transformation of the ER model to SQL

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mappe d CLOs	Teaching Strategies	Assessment Strategies
Week 1	Files and Databases, Database Management systems, Data models	Define Database Explain advantage of Database over file system Describe different data model	CLO 1	Lecture, handouts	Quiz Participation Exam Assignment
Week 2 ,3	Relations, Domains, Attributes and Tuple	Explain Relational Data model Apply RDM in DB design Describe different types of attribute	CLO 2	Lecture, handouts	Quiz Participation Exam Assignment

Week 4, 5	Anomalies, Functional Dependency, First, Second and third normal forms, Boyce-Codd Normal form	Evaluate anomalies in DB design Identify different normal form Apply normalization in DB design Convert one normal form to another normal	CLO 3	Lecture, handouts	Quiz Participation Exam Assignment
Week 6, 7, 8	Relational Calculus Based Languages: SQL, Relational algebra and Set operations	Understanding the concept of SQL	CLO 4	Lecture, handouts	Quiz Participation Exam Assignment
Week 9, 10	Relational design criteria, Lossless decomposition, decomposition algorithms, synthesis algorithms	Explain decomposition and synthesis algorithm Define lossless decomposition	CLO 5	Lecture, handouts	Quiz Participation Exam Assignment
Week 11, 12	The ER model and its constructs, ER modeling in logical database design.	Define entity Relationship, relationship set. Apply E-R approach to DB design	CLO 6	Lecture, handouts	Quiz Participation Exam Assignment
Week 13, 14	Transformation of the ER model to SQL	Explain transformation of ER model to SQL	CLO 7	Lecture, handouts	Quiz Participation Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	Description
Participation	Attendance and participate in the class discussion

Quiz	Quizzes are simple class tests with duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (40 Marks):

Bloom's Category Marks (out of 20)	Test (30)	Attendance (10)	External Participation in Curricular/Co-Curricular Activities
Remember	2		
Understand	5		
Apply	13		
Analyze	2		
Evaluate	4		
Create	4		

SE- Lab Final Exam (60 marks)

Bloom's Category	Test
Remember	5
Understand	10
Apply	25
Analyze	10
Evaluate	5
Create	5

Mapping of CLOs to PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
CLO1	1	0	0	0	0	0	0	1	0
CLO2	1	0	0	0	0	0	0	1	0
CLO3	1	0	0	0	0	0	0	1	0
CLO4	1	0	0	0	0	0	0	1	0
CLO5	1	0	0	0	0	0	0	1	0
CLO6	1	0	0	0	0	0	0	1	0
CLO7	1	0	0	0	0	0	0	1	0

Part D- Resources

Textbooks

1. Database System Concepts by Korth and Silverchatz
2. Principle of Database Systems by O. William

Reference Books

1. Relational Database Management System by Jeffrey Ullman

Other resources (Slides)

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering

Course Outline: Digital System Design

Part A- Introduction

I. Course code: CSE 2205

II. Credit: 3

Register Transfer Logic: Inter Register Transfer, Arithmetic, Logic and Shift Micro-Operations, Conditional Control Statements, Fixed-Point Binary data, Overflow, Arithmetic shifts, Decimal data, Floating-Point data, Non-numeric data, Instruction codes, Design of simple computer.

Processor Logic Design: Processor Organization, Arithmetic Logic Unit, Design of Arithmetic Circuit, Design of Logic Circuit, Design of Arithmetic Logic Unit, Status Register, Design of Shifter, Processor Unit, Design of Accumulator. Design of Multiplier Circuits.

Control Logic Design: Control Organization, Hardwired control, Micro-program Control, Control of Processor Unit, PLA Control, Micro-program Sequencer.

Computer Design: System Configuration, Computer Instructions, Timing and Control, Execution of Instructions, Design of Computer Registers, Design of Control. Register Load and Inter Register Transfer; Bus Buffer and Memory Cycle of Microcomputers.

Computer Design (Simple as Possible): Architecture, Instruction Set, Programming, Fetch Cycle, Execution Cycle.

Recommended Texts:

1. Principle of Digital Electronics. Author: Malvino
2. Digital Principles & System Design, Author: A.P.Godse, D.A.Godse

Faculty of Science and Engineering
Department of Computer Science and Engineering
Course Outline: Software Development Project- 1

Part A- Introduction

III. Course code: CSE 2208

IV. Credit: 1.5

4. Course Summary

This course gives students experience designing, implementing, testing, and debugging large programs. This C#.net /ASP.net/PHP/Java Programming Knowledge is valuable to both beginners and advanced developers that already have experience in developing applications software. Students will also get advanced programming experience in C#.net /ASP.net/PHP/Java; covering topics such as inheritance, multithreading, networking, database programming, and web development.

5. Course Objectives

- Will be able to understand how programming solve real life problems using computer
- Learn to develop software projects that support organization's strategic goals
- Match organizational needs to the most effective software development model
- Plan and manage projects at each stage of the software development life cycle (SDLC)
- Create project plans that address real-world management challenges
- Develop the skills for tracking and controlling software deliverable

6. Course Learning Outcomes

- CLO1. Able to independently design programs
- CLO2. Learn to produce professional-quality code
- CLO3. Implement large programs
- CLO4. Design and execute tests to identify software bugs
- CLO5. Repair software bugs, redesigning and refactoring code when necessary
- CLO6. Utilize, analyze, and critique code written by others

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	Introduction to Software Project, Program execution, Difference between IDE and Compiler, Programming from scratch	Get the idea of the course, Have real life knowledge of program execution and the main difference between IDE and compiler, Learn how to run a program without using any IDE	CLO1, CLO2	Lecture, Demonstration, Practical example	Quiz, Short question, Problem solving
Week 2	Programming and Problem-Solving common sense, Concepts of IDEs (Microsoft Visual Studio, Phpstorm, Eclipse etc.), Giving view to software project	Learn how programming will help you in times of building real life software, Learn how to solve real life problem using computer , Gain the philosophy of programming	CLO1, CLO2, CLO3	Lecture, Demonstration, Practical example	Quiz, Short question, Problem solving
Week 3	Programming and Problem-Solving common sense, Building the base, OOP review	Learn how programming will help you in times of building real life software, Learn how to solve real life problem using computer, Gain the philosophy of programming	CLO1, CLO2, CLO3	Problem-based Learning, Demonstration, Project /Assignment	Group Exercise, Observation, Inventories
Week 4	To Apply OOP Knowledge	Data Types, Type Conversion, Boxing & Unboxing, Conditional Statements, Looping, Methods in C#,	CLO2, CLO3, CLO4	Problem-based Learning, Demonstration	Group Exercise, Observation, Inventories

		Properties, Arrays, Indexers, Structures, Enumerations		tion, Project /Assignment	
Week 5	To Apply OOP Knowledge	Garbage Collector, Stack and Heap, System. GC Class Understand the major qualities of a Good Software, Encapsulation, Inheritance, Polymorphism, Class and Object Constructors, Dynamic types, Optional parameters, Names & optional arguments, Covariant generic type parameters, Destructors, Method overloading Method overriding, Early binding, Late Binding, Abstract Classes, Abstract Methods, Interfaces, Multiple Inheritance, Generic classes, Static classes, Static constructors, Object initialize	CLO2, CLO3, CLO4, CLO5, CLO6	Problem-based Learning, Demonstration, Project /Assignment	Group Exercise, Observation, Inventories
Week 6	To Apply OOP Knowledge	System Defined Exceptions, Custom Exceptions, Try, Catch, Finally, Throwing exceptions, Function Pointers, Multi cast delegates, File Handling System. IO namespace, File stream, Stream Reader, Stream writer, File info, Directory info, Drive Info	CLO2, CLO3, CLO4, CLO5, CLO6	Problem-based Learning, Demonstration, Project /Assignment	Group Exercise, Observation, Inventories
Week 7,8	To Apply .NET	Developing Microsoft.NET Applications for	CLO2, CLO3, CLO4,	Problem-based	Group Exercise, Observation,

	Knowledge	Windows (Visual C#.NET) Creating a Form Adding Controls to a Form Creating an Inherited Form Organizing Controls on a Form Creating MDI Application Working with Controls Creating an Event Handler for Control Using Windows Forms Controls Using Dialog Boxes Application Adding Controls at Run Time Creating Menus Validating User Input Using Data in Windows Forms Applications Adding ADO.NET Objects to and Configuring ADO.NET Objects in a Windows Forms Application Accessing and Modifying Data by Using DataSets Binding Data to Controls Overview of XML Web Services Persisting Data	CLO5, CLO6	Learning, Demonstration, Project /Assignment	Inventories
Week 9	To Apply .NET Knowledge	Developing Microsoft.NET Applications for Windows (Visual C#.NET) Printing and Reporting	CLO2, CLO3, CLO4, CLO5, CLO6	Problem-based Learning, Demonstration,	Group Exercise, Observation, Inventories

		in Windows Forms Applications Lessons Printing From a Windows Forms Application Using the Print Preview, Page Setup, and Print Dialogs Constructing Print Document Content by Using GDI+ Creating Reports by Using Crystal Reports Deploying Windows Forms Applications .NET Assemblies		Project /Assignment	
Week 10	To Apply .NET Knowledge	Introduction to LINQ and ADO.NET Entity Framework. LINQ expressions Using via extension methods, Filtering, Sorting, Aggregation, Skip and Take operators, Joins, Query, Lambda expressions. Data Projection Single result value, Existing types, Anonymous types, Grouping	CLO3, CLO4, CLO5, CLO6	Problem-based Learning, Demonstration, Project /Assignment	Group Exercise, Observation, Inventories
Week 11, 12, 13	To Apply .NET Knowledge	ASP.NET INTRODUCTION Difference Between ASP and ASP.NET ,Architecture Inline Technique & Code-Behind Technique, Code Render Blocks Server Controls ,Page	CLO3, CLO4, CLO5, CLO6	Problem-based Learning, Demonstration, Project	Group Exercise, Observation, Inventories

		Basics, Page lifecycle, Post back Request View State, Directives PROGRAMMING WITH SERVER CONTROLS Web Server Controls Basic Web Controls, List Controls, Data Controls,Adv Controls, User Controls, Master Page and Content Page. Validation Controls Understanding Validation Client or Server Site Validation Required Filed Validator Rang Validator, Regular Expression Validator, Compare Validator, Custom Validator Validator Summary. CONTENT Developing Microsoft.NET Applications for Web (ASP.NET using C#.NET) STATE MANAGEMENT WITH ASP.NET Context, View State, Cookie State Session State, Session Tracking Application Object, Session and Application Events ADO.NET AND		/Assign ment	
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		ASP.NET Working with Data Controls, GridView, -Inserting, Updating, Deleting,-Sorting in Data Grid -Paging in Data Grid, DataSource Controls, Dataset, DetailsView FormView, Data List, Repeater Control, Crystal Reports ADO.NET PROGRAMMING Architecture, DataReaders and DataSets, Command Object Transaction Programming Procedure Execution Data Adapter and Data Set, Data Tables, Data Relation, Data Views Updating Database			
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Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	Description
Quiz, Short questions	These are simple class tests with duration from 10 minutes to 90 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of homework assessment, students are given specific tasks and instructed to complete them within a given

	period of time.
Implementation	This part is the main development part of the project where students will use different Programming languages and tools to develop software.
Report and Viva	Students have to prepare a document including all the requirements, tools, methodologies, coding samples, output scenario snapshots and raw data used in projects and have to submit that in a proper format. Also, Students have to face a viva where he/she have to answer questions regarding the developments of the project.

Mapping of CLOs to Assessment

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (40 Marks):

Bloom's Category Marks (out of 20)	Test (30)	Attendance (10)	External Participation in Curricular/Co-Curricular Activities
Remember	2		
Understand	5		
Apply	13		
Analyze	2		
Evaluate	4		
Create	4		

SE- Lab Final Exam (60 marks)

Bloom's Category	Test
Remember	5
Understand	10
Apply	25
Analyze	10
Evaluate	5

Create	5
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Mapping of CLOs to PLOs

CLOs	PLO10	PLO11	PLO12	PLO13	PLO14	PLO15	PLO16	PLO17	PLO18
CLO1	1	1	0	0	0	0	0	1	0
CLO2	1	1	0	0	0	0	0	1	1
CLO3	1	1	0	0	0	0	0	1	1
CLO4	1	1	0	0	0	0	0	1	1
CLO5	1	1	0	0	0	0	0	1	1
CLO6	1	1	0	0	0	0	0	1	1

Part D- Resources

Textbooks

3. Online resources: w3schools.com, stackoverflow.com, codeforces.com etc.
4. Online tutorial from udemy, youtube.com etc.

Faculty of Science and Engineering
Department of Computer Science and Engineering

Course Outline: Probability and Statistics for Engineers

Part A- Introduction

I. Course code: MATH 2201

II. Credit: 3

1. Course Summary

This is an introductory course on statistics and probability. All the basic methods of statistics and probability will be covered in this course.

2. Course Objectives

1. Receive a basic knowledge of statistics and probability
2. Able to solve basic statistics and probability related problems using a variety of skills and strategies.
3. Able to analyze data and find insights

3. Course Learning Outcomes

CLO1. Devise the primary idea to statistics and probability

CLO2. Learn to solve statistics and probability related problems using programming languages.

CLO3. Design the experiment to understand real life problem

CLO4. Explore different theories of statistics and probability

CLO5. Utilize different Statistical Methods

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	Preliminaries	Getting familiar with different concepts	CLO1	Lecture	Quiz
Week 2, 3	Measures of Central Tendency	Arithmetic Mean, Geometric Mean, Harmonic Mean, Median, Mode, Weighted mean	CLO1, CLO2	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 4, 5, 6	Measures of Dispersion	Range, Standard Deviation, Mean Deviation, Quartile Deviation, Variance, Moments, Skewness and Kurtosis	CLO3	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 7, 8	Correlation Theory	Linear Correlation - Its measures and significance, Rank Correlation	CLO1	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 9, 10	Regression Analysis	Linear and non-linear regression, Least-square method of curve fittings	CLO1, CLO2, CLO5	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 10, 11	Probability	Elementary concepts, Laws of probability, Conditional Probability and Bay's theorem, Random variables	CLO5	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 12, 13, 14	Probability Distributions	Binomial distribution, Poisson distribution and Normal distribution	CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	Description
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific tasks and instructed to complete them within a given period of time.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	Test (10)	Assignment (5)	Quizzes (5)	External Participation in Curricular/Co-Curricular Activities
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Understand	5			
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SEE-Semester End Examination (72 Marks)

Bloom's Category	Test
Remember	05
Understand	15
Apply	20

Analyze	10
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Mapping of CLOs to PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
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CLO3	1	1	0	0	0	0	0	1	1
CLO4	1	1	0	0	0	0	0	1	1
CLO5	1	1	0	0	0	0	0	1	1

Part D- Resources

Textbooks

1. Mathematical Statistics, Author: Kapur, J.N. and Saxena, H.C.
2. A First Course in Mathematical Statistics, Author: Weatherburn, C.E.

Reference Books

1. Probability and Mathematical Statistics, Author: MarekFisz
2. Fundamentals of Mathematical Statistics, Author: Gupta, S.C. and Kapoor V.K.

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering

Course Outline: Linear Algebra and Fourier Analysis

Part A- Introduction

I. Course code: MATH 2203

II. Credit: 2

1. Course Summary

One of the main goals this course is to establish rules for the limiting behavior of functions so that we can deal with functions with as much confidence as we do real or complex numbers. An equally important motivation (that will only become clear in the second half) is that the systematic study of Fourier series requires the Lebesgue integral. The square mean convergence of Fourier series and Parseval's formula cannot be stated accurately in proper generality without the Lebesgue integral and Lebesgue integrable functions.

2. Course Objectives

1. account for basic concept and theorems within the Linear Algebra and Fourier analysis;
2. demonstrate basic numeracy skill concerning the concepts in the previous point; use the numeracy skill at the solution of mathematical and physical problems
3. formulated as ordinary or partial differential equations.

3. Course Learning Outcomes

- CLO1.** Explain basic concepts of Matrices and their significance in engineering field.
- CLO2.** Understand and explain basic concepts of systems of linear equations.
- CLO3.** Learn determinant of a 2x2 Matrix, determinant of a 3x3 matrix.
- CLO4.** Understand and describe importance of Inverse in solving Linear Systems,
- CLO5.** Explain scalars and vectors, geometrical representation of Vectors

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mappe d CLOs	Teaching Strategies	Assessme nt Strategies
Week 1 Week 2 Week 3	Matrices and their Significance, Matrix Notation, Dimension (Order) of a Matrix, Addressing Elements of a Matrix, Solving Linear Systems in 2 Unknowns, Types of Matrices, Addition and Subtraction of Matrices, Multiplication of Scalars with Matrices, Multiplication of two Matrices	Matrices	CLO1	Lecture Exercise Demonstra tion	Quiz Assignme nt Practical Exam
Week 4 Week 5 Week 6	Systems of Linear Equations Preview, Elementary Row Operations, Row Echelon Form (REF) Determinant of a 2x2 Matrix, Determinant of a 3x3 Matrix, Finding Determinants Quickly Inverse exists only for Square Matrices, Singular Matrices, Importance of Inverse in solving Linear Systems, Inverse of a 2x2 Matrix, Inverse of a 3x3 Matrix - The Two Methods	Linear Equations	CLO1	Lecture Exercise Demonstra tion	Quiz Assignme nt Practical Exam
Week 7 Week 8	Scalars and Vectors, Geometrical Representation of Vectors, Vector Addition and Subtraction, Laws of Vector Addition and Head to Tail Rule, Unit Vector Introduction to Vector Spaces, Euclidean Vector Spaces - Part 1, Euclidean Vector Spaces - Part 2,	Vector Spaces	CLO4	Lecture Exercise Demonstra tion	Quiz Assignme nt Practical Exam

	Euclidean Vector Spaces - Part 3, Definition and Closure Properties, Axioms of Vector Spaces, Subspace and Null space.				
Week 9 Week 10	Fourier series, Fourier coefficients, trigonometric polynomials and orthogonality.	Fourier series	CLO4	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 11 Week 12	Properties of Fourier coefficients; Bessel's inequality, Parseval's identity and the Riemann-Lebesgue lemma.	Fourier coefficients	CLO3	Lecture Exercise Demonstration	Quiz Assignment Practical Exam
Week 13 Week 14	Various notions of convergence of Fourier series, including pointwise, uniform and mean square convergence. Summability methods, convolution and Young's inequality. - Fourier Analysis in broader contexts; for example, Fourier integrals, Fourier expansions in groups, Schwartz spaces and tempered distributions.	convergence of Fourier series	CLO2, CLO5	Lecture Exercise Demonstration	Quiz Assignment Practical Exam

Part C- Assessment and Evaluations

Assessment Procedures

Assessment Name	Description
Quiz	Quizzes are simple class tests with a duration from 20 minutes to 120 minutes. It can be online or offline, students need to prepare for the quiz based on the instructed syllabus.
Assignment	Type of home work assessment, students are given specific

	tasks and instructed to complete them within a given period of time.
Practical	Mostly related to laboratory works or by implementing in real code in either exercise book or in programming IDE
Exam	Each course contains a final exam considering the complete syllabus. It should be a 3hours exam for 72 marks. Students need to answer at most 6 questions. Every question may include sub questions

Mapping of CLOs to Assessment

	Quiz (10%)	Assignment (8%)	Practical (10%)	Final (72%)	Quiz (10%)
CLO1	1	1	1	1	1
CLO2	1	1	1	1	1
CLO3	1	1	1	1	1
CLO4	1	1	1	1	1
CLO5	1	1	1	1	1

Mapping of CLOs to PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
CLO1	1	0	0	0	0	1	0	0	0
CLO2	0	1	0	0	0	0	1	0	0
CLO3	0	0	1	0	0	0	0	1	1
CLO4	0	0	0	1	0	0	0	1	0
CLO5	0	0	0	0	1	0	0	0	1

Part D- Resources

H. Anton and C.Rorres

Brestscher

Lipschutz, S.

Haward A.

Haward A.

: Linear Algebra with Applications.

: Linear Algebra with Applications.

: Linear Algebra

: Elementary Linear Algebra with Application

: Contemporary Linear Algebra

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering
Course Outline: Business Psychology

Part A- Introduction

I. Course code: HUM 2201

II. Credit: 2

1. Course Summary

This course is designed to provide a broad overview of the field of Psychology. Special attention will be given to helping the student become a better thinker by learning cognitive science, industrial psychology, and using existing knowledge to draw conclusions, make predictions, or construct explanations. The goal of this course is to think consciously, deliberately and skillfully about human behavior and hence, the topics such as physiological psychology, object recognition and language understanding, job analysis, method of section, learning, cognition, emotions, as well as others are included. The concept of behavior-based safety program and insights into accident prevention are included in this course.

2. Course Objectives

The objectives of this course is to give students the idea of the discussing the research and theory as it relates to the following topics such as personnel, employee motivation and satisfaction, group processes and leadership, and organizational change and development, self-efficacy and self-safety and describing the scope of study in the field of industrial and organizational psychology and the explanation of measurement and determinants of job satisfaction.

3. Course Learning Outcomes

CLO1. Analyze the basic theories of motivation, including concepts such as instincts, drive reduction, and self-efficacy, self- safety and apply the nature of cognitive science, introduce business psychology.

CLO2 Understand the theory suggested for organizational development and create the role of learning industrial theory in the workplace.

CLO3. Analyze the types and causes of accidents with diagram and extending cooperation concerning industrial safety measures.

CLO4. Apply different OOP design paradigms like use case diagram, sequence diagram, ER diagram and data flow diagram.

CLO5. Apply engineering design to produce solutions that meet specified needs with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors

CLO6. Evaluate the ability to think critically, to analyze complex and diverse concepts, and to use reason and judgment and explain the measurement and determinants of job satisfaction.

Part B- Lesson Plan

Course Details Learning plan

Timeline	Topics /contents	Learning Outcomes	Mapped CLOs	Teaching Strategies	Assessment Strategies
Week 1	Introduction to Psychology, business psychology.	-Student will able to explain the history and methods of psychology, concepts of business psychology.	CLO1	Lecture	Quiz

Week 2, 3	Cognitive Science.	-students will be able to understand the cross disciplinary, historical foundations of cognitive science. -understand the research problems that unify the cognitive sciences, including language, thought, perception, attention, learning, memory, reasoning, problem solving, judgment, and decision-making. -Understand the assumptions, methods and concepts that unify and differentiate the cognitive sciences. -Understand the diversity of theoretical approaches and paradigms across the cognitive sciences, including neural, embodied, social, and/or technological approaches.	CLO1	Lecture, Exercise Demonstration	Quiz, Assignment, Exam
Week 4	Reasoning.	-students will be able to explain the three methods of reasoning which are the deductive, inductive, and abductive approaches.	CLO5	Lecture, Exercise Demonstration	Quiz, Assignment, Exam

Week 5, 6	Object recognition and language understanding. Learning Industrial Psychology	-student will able to give the explanation of object recognition and observe the data spatial resolution. -understand and appreciate the role of human behavior, -Demonstrate knowledge of the biological and conceptual languages of the brain and their potential. -understand and appreciate the connection between scientific inquiry and the creative and artistic dimensions in the field of Psychology	CLO2, CLO4	Lecture, Exercise Demonstration	Quiz, Assignment, Exam
Week 7, 8, 9	Introduction to Job and Job analysis, Methods of selection.	-students will able to understand about Job analysis which is a procedure through which they determine the duties and responsibilities, nature of the jobs and finally to decide qualifications, skills and knowledge to be required for an employee to perform particular job. -understand what tasks are important and how they are carried on. Job analysis forms basis for later HR activities such as developing effective training program, selection of employees, setting up of performance standards and assessment of employee's performance appraisal and employee remuneration system or compensation plan.	CLO5	Lecture, Exercise Demonstration	Quiz, Assignment, Exam

Week 10	Training in Industry	-student will able to know about coaching, training program implementation, instructional design, and management training.	CLO2, CLO3	Lecture, Exercise Demonstration	Quiz, Assignment, Exam
Week 11, 12	Motivation and Work, Job satisfaction	-Students will able to understand about Illustrate intrinsic and extrinsic motivation and describe basic theories of motivation, including concepts such as instincts, drive reduction, and self-efficacy. -explain the measurement and determinants of job satisfaction.	CLO1, CLO5	Lecture, Exercise Demonstration	Quiz, Assignment, Exam
Week 13, 14	Introduction to Ergonomics, System Engineering , Accident and Safety.	- Students will able to understand the basic physiological systems, nervous system, respiratory system, circulatory system, metabolic system. -analyze the causes of a systems failure. -answer the question 'why is systems engineering important? -explain types and causes of accidents with diagram and extending cooperation concerning industrial safety measures.	CLO1, CLO4	Lecture, Exercise Demonstration	Quiz, Assignment, Exam

Part C- Assessment and Evaluations

Assessment Procedures

Mapping of CLOs to Assessment

CIE- Continuous Internal Evaluation (28 Marks):

Bloom's Category Marks (out of 20)	Test (10)	Assignment (5)	Quizzes (5)	External Participation in Curricular/Co-Curricular Activities
Remember	2			
Understand	5			
Apply	3			
Analyze	2			
Evaluate	4			
Create	4			

SEE-Semester End Examination (72 Marks)

Bloom's Category	Test
Remember	05
Understand	15
Apply	20
Analyze	10
Evaluate	10
Create	12

Mapping of CLOs to PLOs

CLOs	PLO1	PLO2	PLO3	PLO4	PLO5	PLO6	PLO7	PLO8	PLO9
CLO1	1	1	1	0	0	0	0	1	1
CLO2	1	1	1	0	0	0	0	1	0
CLO3	1	1	1	0	0	0	0	1	0
CLO4	1	1	1	0	0	0	0	1	1

CLO5	1	1	1	0	0	0	0	1	1
CLO6	1	1	1	0	0	0	0	1	1

Part D- Resources

TEXT BOOKS:

- 1.* A Text-Book of Psychology by Edward Bradford Titchener.
- 2.* The Social Animal by Elliot Aronson.

REFERENCE BOOKS:

- 1.* Authentic Happiness by Martin Seligman
- 2.* The Happiness Hypothesis by Jonathan Heidt
- 3.* Influence: The Psychology Of Persuasion by Robert B. Cialdini

Jashore University of Science and Technology
Faculty of Science and Engineering
Department of Computer Science and Engineering

Course Outline: Viva Voce

Part A- Introduction

- I. Course code: CSE 2200**
- II. Credit: 1**